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Current Position

Korea Institute for Advanced Study (KIAS) — Seoul, Republic of Korea
AI Fellow, Center for AI and Natural Sciences — *March 2026–present*

Previous Position

Institute for Basic Science (IBS ECOPRO) — Daejeon, Republic of Korea
Student Researcher, Extremal Combinatorics and Probability Group — *April 2022–February 2026*

Education

KAIST — Daejeon, Republic of Korea
Ph.D. in Mathematical Sciences — *March 2021–February 2026*

- Degree conferred on February 20, 2026.
- Advised by Jaehoon Kim and Hong Liu.

KAIST — Daejeon, Republic of Korea
B.S., Department of Mathematical Sciences — *March 2016–August 2020*

Hansung Science High School (한성과학고등학교) — Seoul, Republic of Korea
High school education — *March 2014–February 2016*

Research Interests

- Extremal and probabilistic combinatorics, especially
  - Dirac-type problems;
  - spanning structures in random and randomly perturbed graphs and hypergraphs;
  - graph transversals and rainbow structures;
  - graphon optimization and homomorphism counting.
- Computational complexity of graph problems.
- AI-assisted approaches to combinatorics.

Published and Accepted Articles

Unless otherwise indicated, papers follow the alphabetical authorship convention in mathematics.

To appear

1. **Unbounded degree spanning hypertrees in Dirac hypergraphs** (with Yaobin Chen and Junchi Zhang).
   To appear in *Israel Journal of Mathematics*.
   arXiv:2508.06843.

2026

2. **Sidorenko's conjecture for subdivisions and theta substitutions** (with Ruonan Li and Hong Liu).
Combinatorics, Probability and Computing 35(2) (2026), 269–279.
arXiv:2408.03491; doi:10.1017/S0963548325100242.

2025

3. **The proper conflict-free k-coloring problem and the odd k-coloring problem are NP-complete on bipartite graphs** (with Jungho Ahn and Sang-il Oum).
Discrete Applied Mathematics 377 (2025), 10–17.
arXiv:2208.08330; doi:10.1016/j.dam.2025.06.026.

4. **On rainbow Turán densities of trees** (with Jaehoon Kim, Hyunwoo Lee, and Haesong Seo).
Random Structures & Algorithms 66(3) (2025), e70005.
arXiv:2312.15956; doi:10.1002/rsa.70005.

5. **Dirac's theorem for linear hypergraphs** (with Hyunwoo Lee).
SIAM Journal on Discrete Mathematics 39(2) (2025), 834–847.
arXiv:2403.14269; doi:10.1137/24M1659467.

6. **Crux, space constraints and subdivisions** (with Jaehoon Kim, Younjin Kim, and Hong Liu).
Journal of Combinatorial Theory, Series B 170 (2025), 82–127.
arXiv:2207.06653; doi:10.1016/j.jctb.2024.08.005. An extended abstract appeared in EUROCOMB'23.

2024

7. **A proof of the Elliott–Rödl conjecture on hypertrees in Steiner triple systems** (with Jaehoon Kim, Joonkyung Lee, and Abhishek Methuku).
Forum of Mathematics, Sigma 12 (2024), e75.
arXiv:2208.10370; doi:10.1017/fms.2024.34.

8. **On the spectral radius of graphs with given maximum degree and girth** (with Jiangdong Ai, Jaehoon Kim, Hyunwoo Lee, Suil O, and Liwen Zhang).
Linear Algebra and its Applications 691 (2024), 182–195.
doi:10.1016/j.laa.2024.03.026.

2016

9. **On the mean square displacement of a random walk on a graph** (with Hwidong Kim, Jiho Maeng, Jihwan Yu, Yongwook Cha, and Seong-Hun Paeng).
European Journal of Combinatorics 51 (2016), 227–235.
doi:10.1016/j.ejc.2015.05.009.

Preprints

1. **Goodman-type lower bounds for odd cycles I: The dense regime** (with Taeyoung Kim and Hongseok Yang).
arXiv:2610.18277 (2026).

2. **A new upper bound for the Turán density of the tetrahedron** (with Gyeongwon Jeong, Seonghun Park, Joonkyung Lee, and Hongseok Yang; joint first authors: Gyeongwon Jeong and Seonghun Park).
arXiv:2609.27495 (2026).

3. **Bipodal optimizers in the upper-tail variational problem for regular subgraph densities** (with Sangho Lim, Taeyoung Kim, Kyeongsik Nam, and Hongseok Yang; first author: Sangho Lim).
arXiv:2609.15222 (2026).

4. **On the enumeration of polymatroids** (with Donggyu Kim).
arXiv:2608.18225 (2026).

5. **Note on the codegree version of the Erdős–Ko–Rado theorem** (with Luyining Gan and Jie Han).
arXiv:2605.17945 (2026).

6. **Hitting time for Hamilton cycles in pseudorandom graphs** (with Yaobin Chen, Yu Chen, and Yiting Wang).
arXiv:2603.05269 (2026).

7. **The perturbation threshold of degenerate graphs** (with Jie Han, Bin Wang, and Junxue Zhang).
arXiv:2602.21867 (2026).

8. **Berge Hamilton cycles in a random sparsification of dense hypergraphs** (with Minseo Kim).
arXiv:2512.06675 (2025).

9. **An efficient algorithm for $\mathcal{F}$-subgraph-free edge deletion on graphs having a product structure** (with Shinwoo An, Seokbeom Kim, and Myounghwan Lee).
arXiv:2510.14674 (2025).

10. **Perturbation of dense graphs** (with Jie Han, Bin Wang, and Junxue Zhang).
arXiv:2508.18042 (2025).

11. **On high discrepancy 1-factorizations of complete graphs** (with Jiangdong Ai, Fankang He, and Hyunwoo Lee).
arXiv:2503.17176 (2025).

12. **Ramsey–Dirac theory for bounded degree hypertrees** (with Jie Han, Jaehoon Kim, and Donglei Yang).
arXiv:2411.17996 (2024).

13. **Graph with any rational density and no rich subsets of linear size** (with Suyun Jiang, Hong Liu, and Tuan Tran).
arXiv:2402.13825 (2024).

14. **A bandwidth theorem for graph transversals** (with Debsoumya Chakraborti, Jaehoon Kim, and Hong Liu).
arXiv:2302.09637 (2023).

Other Manuscript

- **Complexity of Partitioning Hypergraphs**. arXiv:1812.09206.

Talks

2026

- **International Conference for the 80th Anniversary of the Korean Mathematical Society – Discrete Mathematics**
June 24, 2026, Seoul, Republic of Korea
The perturbation threshold of degenerate graphs (event page)

- **Hanyang University Mathematics Seminar** *Invited*
  April 27, 2026, Hanyang University, Seoul, Republic of Korea
  Hitting times for Hamilton cycles in random sparsifications (event page)
- **Beijing Institute of Technology Combinatorics Seminar** *Invited*
  April 16, 2026, Beijing Institute of Technology, Beijing, China
  Hitting times for Hamilton cycles in random sparsifications
- **36th KIAS Combinatorics Workshop** *Invited*
  March 14, 2026, Seoul, Republic of Korea
  Embedding loose hypertrees into hypergraphs (event page)

2025

- **2025 Combinatorics Workshop**
  August 18, 2025, Daejeon, Republic of Korea
  Random perturbation of dense graphs (event page)
- **Discrete Analysis at Yonsei** *Invited*
  June 19, 2025, Yonsei University, Seoul, Republic of Korea
  Ramsey–Dirac theory for bounded degree hypertrees (event page)
- **Korean Mathematical Society Spring Meeting**
  April 25, 2025, KAIST, Daejeon, Republic of Korea
  Spanning hypertrees in dense pseudorandom hypergraphs (event page)
- **4th East Asia Workshop on Extremal and Structural Graph Theory** *Invited*
  March 29, 2025, Sun Yat-sen University, Guangzhou, China
  Ramsey–Dirac theory for bounded degree hypertrees (event page)

2024

- **Beijing University of Posts and Telecommunications Workshop** *Invited*
  December 15, 2024, Hainan, China
  Ramsey–Dirac theory for bounded degree hypertrees
- **Beijing Institute of Technology Seminar** *Invited*
  December 10, 2024, Beijing Institute of Technology, Beijing, China
  Almost spanning hypertrees in a Steiner triple system
- **Korean Mathematical Society Annual Meeting – Special Section: Recent Developments in Combinatorics** *Invited*
  October 25, 2024, Sungkyunkwan University, Suwon, Republic of Korea
  Sidorenko's conjecture for theta substitutions (event page)
- **Discrete Analysis Seminar, Yonsei University** *Invited*
  October 15, 2024, Seoul, Republic of Korea
  Rainbow Turán densities of trees via graph limits (event page)
- **Shandong University Seminar** *Invited*
  August 5, 2024, online
  Rainbow Turán densities of trees via graph limits
- **9th European Congress of Mathematics – Combinatorics and Discrete Mathematics**
  July 16, 2024, Seville, Spain
  Graph with any rational density and no rich subsets of linear size (event page)
- **Summit280**
  July 11, 2024, Budapest, Hungary
  Dirac's theorem for linear hypergraphs (event page)

- **30th British Combinatorial Conference**
  July 3, 2024, London, United Kingdom
  Dirac's theorem for linear hypergraphs (event page)
- **31st KIAS Combinatorics Workshop** *Invited*
  June 1, 2024, Jeju, Republic of Korea
  Dirac's theorem for linear hypergraphs (event page)
- **Korean Mathematical Society Spring Meeting**
  April 19, 2024, Daejeon, Republic of Korea
  On rainbow Turán densities of trees (event page)
- **Yeungnam University Combinatorics Seminar** *Invited*
  March 18, 2024, Yeungnam University, Gyeongsan, Republic of Korea
  Graph with any rational density and no rich subsets of linear size (event page)

2023

- **European Conference on Combinatorics, Graph Theory and Applications (EUROCOMB'23)**
  August 31, 2023, Prague, Czech Republic
  Crux, space constraints and subdivisions (extended abstract)
- **Korean Mathematical Society Spring Meeting – Special Section: Extremal Combinatorics: Methods and Applications** *Invited*
  April 29, 2023, Daejeon, Republic of Korea
  A bandwidth theorem for graph transversals (event page)
- **Shandong University Seminar** *Invited*
  March 30, 2023, online
  A bandwidth theorem for graph transversals (video)

2022

- **IBS Discrete Mathematics Seminar** *Invited*
  November 29, 2022, IBS, Daejeon, Republic of Korea
  A proof of the Elliott–Rödl conjecture on hypertrees in Steiner triple systems (video)
- **KAIST Mathematics Graduate Student Seminar**
  November 3, 2022, KAIST, Daejeon, Republic of Korea
  Large clique subdivisions in graphs without small dense subgraphs (event page)

2021

- **2021 Combinatorics Workshop**
  December 21, 2021, Yangpyeong, Republic of Korea
  Large clique subdivisions in graphs without small dense subgraphs (video)
- **IBS Discrete Mathematics Seminar** *Invited*
  November 30, 2021, IBS, Daejeon, Republic of Korea
  Large clique subdivisions in graphs without small dense subgraphs (video)

Teaching

- **Mini-course: Finding Large Structures** (with Hyunwoo Lee)
  Nankai University, February 18–21, 2025.

Teaching Assistantships at KAIST

2025

- Fall: CC511 Probability and Statistics.
- Spring: CS492 Algorithmic Graph Theory.

2024

- Fall: MAS477 Introduction to Graph Theory.
- Spring: MAS275 Discrete Mathematics.

2023

- Fall: MAS102 Calculus II and MAS480 Topological Methods in Combinatorics.
- Spring: MAS101 Calculus I and MAS275 Discrete Mathematics.

2022

- Fall: MAS102 Calculus II and MAS477 Introduction to Graph Theory.
- Spring: MAS102 Calculus II and MAS275 Discrete Mathematics.

2021

- Fall: MAS102 Calculus II and CC511 Probability and Statistics.
- Spring: MAS101 Calculus I (Outstanding Teaching Assistant Award).

Academic Service

Organization

- Co-organizer, ASIACOMB 2026, August 24–28, 2026 (event page).
- Co-organizer, Korean Student Combinatorics Workshop, February 2–6, 2026 (event page).
- Co-organizer, Deep Seminar in Combinatorics, 2023–2025 (event page).
- Co-organizer, Korean Student Combinatorics Workshop, July 29–August 2, 2024 (event page).
- Co-organizer, IBS ECOPRO Student Reading Group, 2022–2023 (event page).

Refereeing

2026 *SIAM Journal on Discrete Mathematics*: 2 manuscripts; *Graphs and Combinatorics*: 1 manuscript.
2025 *Discrete Mathematics*, *Electronic Journal of Combinatorics*, and *SIAM Journal on Discrete Mathematics*: 1 manuscript each.
2024 *Electronic Journal of Combinatorics*: 1 manuscript.
2023 IWOCA 2023: 1 manuscript.

Awards and Honors

- **Noh Yoomi Thesis Award**, 2026 (Korean Mathematical Society).
- **Outstanding Teaching Assistant Award**, 2021 (KAIST).
- **Silver Prize, 37th Mathematical Contest for University Students**, 2018 (Korean Mathematical Society).
- **Fifth Prize (30th place), Kakao Code Festival**, 2018 (Kakao Corporation).
- **Silver Prize, 36th Mathematical Contest for University Students**, 2017 (Korean Mathematical Society).

Research Visits

- Beijing Institute of Technology, Beijing, China, invited by Prof. Jie Han, April 15–21, 2026.
- Nankai University, Tianjin, China, invited by Prof. Jiangdong Ai, February 13–25, 2025.
- Beijing University of Posts and Telecommunications, Hainan Campus, China, invited by Prof. Luyining Gan, December 11–23, 2024.
- Beijing Institute of Technology, Beijing, China, invited by Prof. Jie Han, December 9–11, 2024.
- Nankai University, Tianjin, China, invited by Prof. Jiangdong Ai, February 16–25, 2024.

References

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